Binary Lossy Coding Problem with Encoder Side Information and Common Reconstruction Constraint

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***Abstract*—The source coding problem with common reconstruction constraint is considered, where the encoder can produce an exact copy of the compressed source reconstructed by the decoder. In this paper, we consider the variation of this problem in which the encoder can observe the side information which may be different from the decoder side information. For the doubly-symmetric binary source, we derive the rate distortion function, and perform the utility of encoder side information.**

I. Introduction

The lossy source coding problem with side information has been investigated by Wyner and Ziv [1] in which they showed the rate distortion function when only decoder has access to the side information, where the side information is exploited to not only reduce the information rate but also reduce the reconstruction distortion. Gray [2] [3] computed the conditional rate distortion function when the side information is observed at both the encoder and the decoder. When the encoder and the decoder have individual side information, the rate distortion for the binary source is characterized by [4]. Laich and Wigger [5] discussed the utility of encoder side information on reducing the required coding rate for the lossless source coding problem.

Recently, Steinberg [6] introduced the common reconstruction (CR) constraint into Wyner-Ziv problem, in which the encoder is required to produce the copy of the decoder's reconstruction with probability near one. Based on Steinberg's work, Lapidoth, Malar and Wuigger [7] presented the rate distortion function under a modified constraint on encoder which can compute the decoder's reconstruction within a given distortion constraint. Vellambi and Timo [8] [9] considered a modified Wyner-Ziv setting for two decoders under a common receiver reconstruction constraint which requires that the reconstructions of two receivers are identical in probability near one. For this setting, they characterized the rate region for the successive refinement problem [8] and the rate distortion function for the Heegard-Berger problem [9], respectively.

In this paper, we consider the extension of the source coding problem with CR constraint, where the encoder have the side information which is correlated with the decoder side information. In this setup, the encoder side information may make the encoder easier to estimate the decoder's construction. To discuss the utility of encoder side information, we consider the doubly-symmetric binary source (DSBS) and character the rate distortion function. The remainder of this paper is organized as follows. In Section II, we define the system model. The characterization of the rate distortion for DSBS is provided in Section III. The proof details and conclusions are given in Section IV and V, respectively.

II. System Model

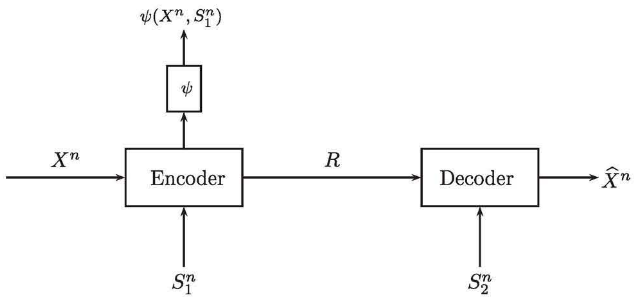

Fig. 1. The lossy source coding problem with encoder side information and common reconstruction constraint.

The model considered in this paper is defined by finite discrete alphabets $\mathcal{X}, \mathcal{S}_1, \mathcal{S}_2, \widehat{\mathcal{X}}$ and the probability mass function (pmd) $p_{XS_1S_2}(x, s_1, s_2)$ as follows. The source $X^n \in \mathcal{X}^n$, side information $S_1^n \in \mathcal{S}_1^n$ and $S_2^n \in \mathcal{S}_2^n$ are such that the tuples (X_i, S_{1i}, S_{2i}) are independent and identically distributed according to joint pmd $p_{XS_1S_2}(x, s_1, s_2)$. In Fig. 1, source sequence X^n and side information S_1^n are observed at the encoder and then encoded into a message in nR bits, which is transmitted to the decoder, which wants to reconstruct the source sequence X^n as $\widehat{X}^n$ under the given distortion constraint. In detail, the decoder can use both the message and the side information S_2^n to obtain $\widehat{X}^n$. The distortion measure between two vectors is defined by a per-letter distortion measure $\mathbf{d}(x, \widehat{x}) : \mathcal{X} \times \widehat{\mathcal{X}} \to [0, \infty)$, from which, the distortion measure between source x^n and estimation $\widehat{x}^n$ is defined as

$$\mathbf{d}_n(x^n, \widehat{x}^n) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{d}(x_i, \widehat{x}_i). \tag{1}$$

Definition 1: An (n, R, D, ϵ) code for the lossy coding problem with encoder side information and common reconstruction constraints consists of an encoding mapping

$$g : \mathcal{X}^n \times \mathcal{S}_1^n \to \mathcal{T} = \{1, 2, \ldots, 2^{nR}\}, \tag{2}$$

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a decoding mapping for the decoder

$$h : \mathcal{T} \times \mathcal{S}_2^n \to \widehat{\mathcal{X}}^n, \tag{3}$$

and a reconstruction mapping at the encoder

$$\psi : \mathcal{X}^n \times \mathcal{S}_1^n \to \widehat{\mathcal{X}}^n, \tag{4}$$

such that the distortion constraint is satisfied, i.e.,

$$\mathbb{E}\mathbf{d}_n(X^n, h(g(X^n, S_1^n), S_2^n)) \leq D, \tag{5}$$

and the CR constraint holds, i.e.,

$$P(\psi(X^n, S_1^n) \neq h(g(X^n, S_1^n), S_2^n)) \leq \epsilon. \tag{6}$$

Rate R is said to be D-achievable, if for any $\epsilon > 0$ and sufficiently large n, there exists an $(n, R, D + \epsilon, \epsilon)$ code. Then, the rate-distortion function $\widetilde{R}^{CR}(D)$ is defined as the minimum rate of all D-achievable rates.

When there is no side information S_1^n in Fig. 1, the rate distortion function is derived in [6]. By setting (X^n, S_1^n) as the source of encoder in the model of [6], the rate distortion function $\widetilde{R}^{CR}(D)$ in Fig. 1 can be derived as follow.

Theorem 1:

$$\widetilde{R}^{CR}(D) = \min_{\widetilde{\mathcal{D}}(D)} I(X, S_1; W|S_2), \tag{7}$$

where $\widetilde{\mathcal{D}}(D)$ is the set of all random variables W satisfy the following conditions: (1) $W \leftrightarrow (X, S_1) \leftrightarrow S_2$ forms a Markov chain; (2) there exists deterministic function f such that $\mathbb{E}\mathbf{d}(X, f(W)) \leq D$.

III. Result for Binary Source

We will highlight the rate distortion function in (7) for the DSBS with Hamming distortion $\mathbf{d}(x, \widehat{x}) = x \oplus \widehat{x}$, where $\oplus$ denotes modulo two addition. The DSBS X is a discrete memeryless source with alphabet in $\{0, 1\}$, and $P(X = 0) = P(X = 1) = 1/2$. Assume the encoder side information S_1 and the decoder side information S_2 are generated as follows,

$$\begin{cases} S_1 = X \oplus N_1, \\ S_2 = S_1 \oplus N_2, \end{cases} \tag{8}$$

where N_1 is a Bernoulli random variable independent of all other variables with $P(N_1 = 1) = p_1 < 1/2$, and N_2 is a Bernoulli random variable independent of all other variables with $P(N_2 = 1) = p_2 < 1/2$. Assumption in (8) means that $X \leftrightarrow S_1 \leftrightarrow S_2$ forms a Markov chain. This assumption is from [4], in which their result shows that the encoder side information is useful for the lossy coding problem without CR constraint. Other cases of Markov chain assumption on the source and the side information will be discussed in [10].

Now, we need to recall the following function from [1],

$$G_p(u) \triangleq h(p * u) - h(u), \tag{9}$$

where $u \in [0, 1]$, for given $p \in (0, 1/2)$, $h(u)$ is the binary entropy function $h(u) = -u \log u - (1 - u) \log(1 - u)$ with $h(0) = h(1) = 0$, and $p * u = p(1 - u) + (1 - p)u$ is the binary convolution. Using notion $\overline{m} = 1 - m$, we will define $F(\alpha, \beta, d)$ as below,

$$\begin{aligned} &F(\alpha, \beta, d) \triangleq h(p_1) + G_{p_2}(\alpha\overline{d} + \beta d) \\ &-(\overline{\alpha}\overline{d} + \overline{\beta}d) h(\frac{\overline{\beta}d}{\overline{\alpha}\overline{d} + \overline{\beta}d}) - (\alpha\overline{d} + \beta d) h(\frac{\beta d}{\alpha\overline{d} + \beta d}), \end{aligned} \tag{10}$$

where parameters satisfy $0 \leq \alpha, \beta, d \leq 1$.

Theorem 2:

$$\widetilde{R}^{CR}(D) = \begin{cases} \min\limits_{\substack{0 \leq \alpha, \beta \leq 1, \\ \alpha\overline{D} + \beta D = p_1}} F(\alpha, \beta, D), & 0 \leq D < 1/2, \\ 0, & 1/2 \leq D \leq 1. \end{cases} \tag{11}$$

IV. Proof Details

In this section, we will introduce the following lemma, which will be used to simplify the proof of the Theorem 2.

Lemma 1: When $W \leftrightarrow (X, S_1) \leftrightarrow S_2$ forms a Markov chain, then we have

$$\begin{aligned} I(X, S_1; W|S_2) &= H(X, S_1) - H(S_2) \\ &+ H(S_2|W) - H(S_1|W) - H(X|W, S_1). \end{aligned} \tag{12}$$

The proof of this lemma is given in Appendix.

When $1/2 \leq D \leq 1$, the optimal rate is zero since the source is DSBS. For the case of $0 \leq D < 1/2$, the proof will be divided into two steps. The first step will show

$$\widetilde{R}^{CR}(D) = \min_{\substack{0 \leq \alpha, \beta \leq 1, \\ \alpha\overline{d} + \beta d = p_1, \\ 0 \leq d \leq D}} F(\alpha, \beta, d). \tag{13}$$

In the second step, we will prove the constraint $d \leq D$ can be met with equality without loss of optimality.

Step 1: we prove the (13).

Achievability of $\widetilde{R}^{CR}(D)$ in (13): Consider the distribution $P(w, x, s_1, s_2) = P(w, x, s_1)P(s_2|s_1)$, where $P(w, x, s_1)$ is given in Table I. Using this distribution and Lemma 1, we can derive

$$I(X, S_1; W|S_2) = F(\alpha, \beta, d).$$

Since the distortion measure is Hamming distance, by setting the reconstruction function as $f(W) = W$, we have

$$\mathbb{E}\mathbf{d}(X, W) = \mathbb{E}(X \oplus W) = d. \tag{14}$$

From (13), we have $d \leq D$ which means that the above expected distortion is satisfied the distortion constraint.

	$S_1 = 0$		$S_1 = 1$	
	$X = 0$	$X = 1$	$X = 0$	$X = 1$
$W = 0$	$\frac{\overline{\alpha}\overline{d}}{2}$	$\frac{\overline{\beta}d}{2}$	$\frac{\alpha\overline{d}}{2}$	$\frac{\beta d}{2}$
$W = 1$	$\frac{\beta d}{2}$	$\frac{\alpha\overline{d}}{2}$	$\frac{\overline{\beta}d}{2}$	$\frac{\overline{\alpha}\overline{d}}{2}$

TABLE I

Joint distribution $P(X, S_1, W)$, where parameters satisfy $\alpha\overline{d} + \beta d = p_1$, and the notion $\overline{m} = 1 - m$.

Lower bound of $\widetilde{R}^{CR}(D)$ in (13):

For any $W \in \widetilde{\mathcal{D}}(D_1, D_2)$ considered in (7), they satisfy two conditions: (1) $W \leftrightarrow (X, S_1) \leftrightarrow S_2$ forms a Markov chain; (2) there exists deterministic function f such that $\mathbb{E}\mathbf{d}(X, f(W)) \leq D$. Our goal is to show that

$$I(X, S_1; W|S_2) \geq F(\alpha, \beta, d), \tag{15}$$

from which, the lower bound of the rate distortion function $\widetilde{R}^{CR}(D)$ can be established.

Before starting to bound this rate distortion, we define the following quantities for each w,

$$\begin{aligned} d_w &\triangleq P(X \neq f(W)|W = w), &(16)\\ \alpha_w &\triangleq P(S_1 \neq f(W)|X = f(w), W = w), &(17)\\ \beta_w &\triangleq P(S_1 \neq f(W)|X \neq f(w), W = w). &(18) \end{aligned}$$

To simplify the notions, we write $P(W = w)$ as P_w. Using (16), (17) and (18), we define

$$\begin{aligned} d &\triangleq \sum_w P_w d_w, &(19)\\ \alpha &\triangleq \sum_w \frac{P_w \overline{d_w} \alpha_w}{\overline{d}}, &(20)\\ \beta &\triangleq \sum_w \frac{P_w d_w \beta_w}{d}. &(21) \end{aligned}$$

From Lemma 1, we have

$$\begin{aligned} &I(X, S_1; W|S_2) = H(X, S_1) - H(S_2)\\ &\quad + H(S_2|W) - H(S_1|W) - H(X|W, S_1), \end{aligned} \tag{22}$$

where, the terms $H(X, S_1)$ and $H(S_2)$ can be derived by using assumptions in (8),

$$\begin{aligned} H(X, S_1) &= H(X) + H(S_1|X) = 1 + h(p_1), &(23)\\ H(S_2) &= 1. &(24) \end{aligned}$$

The term $H(S_2|W) - H(S_1|W)$ is calculated as follows.

$$\begin{aligned} &H(S_2|W) - H(S_1|W)\\ =\ &\sum_w P_w[H(S_2|W = w) - H(S_1|W = w)]\\ =\ &\sum_w P_w[h(P(S_2 \neq f(w)|W = w))\\ &-h(P(S_1 \neq f(w)|W = w))]\\ \overset{(a)}{=}\ &\sum_w P_w[h(p_2 * P(S_1 \neq f(w)|W = w))\\ &-h(P(S_1 \neq f(w)|W = w))]\\ \overset{(b)}{=}\ &\sum_w P_w G_{p_2}(P(S_1 \neq f(w)|W = w))\\ \overset{(c)}{\geq}\ &G_{p_2}\left(\sum_w P_w P(S_1 \neq f(w)|W = w)\right)\\ \overset{(d)}{=}\ &G_{p_2}(\alpha\overline{d} + \beta d), \end{aligned} \tag{25}$$

where (a) follows from the following result by using the fact that $W \leftrightarrow S_1 \leftrightarrow S_2$ is a Markov chain,

$$\begin{aligned} &P(S_2 \neq f(w)|W = w)\\ =\ &P(S_2 \neq f(w), S_1 = f(w)|W = w)\\ &+P(S_2 \neq f(w), S_1 \neq f(w)|W = w)\\ =\ &P(S_2 \neq f(w)|S_1 = f(w))P(S_1 = f(w)|W = w)\\ &+P(S_2 \neq f(w)|S_1 \neq f(w))P(S_1 \neq f(w)|W = w)\\ =\ &p_2(1 - P(S_1 \neq f(w)|W = w))\\ &+(1 - p_2)P(S_1 \neq f(w)|W = w)\\ =\ &p_2 * P(S_1 \neq f(w)|W = w), \end{aligned} \tag{26}$$

(b) from the definition of function G in (9), (c) from the convexity of function G in [1], and (d) from the following result by using the definitions between (16)-(21),

$$\begin{aligned} &\sum_w P_w P(S_1 \neq f(w)|W = w)\\ =\ &\sum_w P_w[P(S_1 \neq f(w)|W = w, X = f(w))P(X = f(w)|W = w)\\ &+P(S_1 \neq f(w)|W = w, X \neq f(w))P(X \neq f(w)|W = w)]\\ =\ &\sum_w P_w[\alpha_w(1 - d_w) + \beta_w d_w]\\ =\ &\alpha\overline{d} + \beta d, \end{aligned} \tag{27}$$

Next, we calculate the last term $H(X|W, S_1)$.

$$\begin{aligned} &H(X|W, S_1)\\ =\ &\sum_{w,s_1} P(w, s_1)H(X|W = w, S_1 = s_1)\\ =\ &\sum_{w,s_1} P(w, s_1)h(P(X \neq f(w)|W = w, S_1 = s_1))\\ =\ &\sum_w P_w P(S_1 \neq f(w)|w)h(P(X \neq f(w)|w, S_1 \neq f(w)))\\ +\ &\sum_w P_w P(S_1 = f(w)|w)h(P(X \neq f(w)|w, S_1 = f(w)))\\ \overset{(e)}{\leq}\ &(\alpha\overline{d} + \beta d)\\ &\cdot h\left(\frac{\sum_w P_w P(S_1 \neq f(w)|w)P(X \neq f(w)|w, S_1 \neq f(w))}{\alpha\overline{d} + \beta d}\right)\\ &+(\overline{\alpha}\overline{d} + \overline{\beta}d)\\ &\cdot h\left(\frac{\sum_w P_w P(S_1 = f(w)|w)P(X \neq f(w)|w, S_1 = f(w))}{\overline{\alpha}\overline{d} + \overline{\beta}d}\right)\\ \overset{(f)}{=}\ &(\alpha\overline{d} + \beta d)h(\frac{\beta d}{\alpha\overline{d} + \beta d}) + (\overline{\alpha}\overline{d} + \overline{\beta}d)h(\frac{\overline{\beta}d}{\overline{\alpha}\overline{d} + \overline{\beta}d}), \end{aligned} \tag{28}$$

where (e) follows from concavity of the binary entropy function, the equality in (27) and the following result

$$\begin{aligned} &\sum_w P_w P(S_1 = f(w)|w)\\ =\ &1 - \sum_w P_w P(S_1 \neq f(w)|w)\\ =\ &\overline{\alpha}\overline{d} + \overline{\beta}d, \end{aligned}$$

(f) from the two result as below,

$$\begin{aligned}
& \sum_w P_w P(S_1 \neq f(w)|w) P(X \neq f(w)|w, S_1 \neq f(w)) \\
= & \sum_w P(W = w, S_1 \neq f(w), X \neq f(w)) \\
= & \sum_w P_w d_w \beta_w \\
= & \beta d \qquad (29)
\end{aligned}$$

$$\begin{aligned}
& \sum_w P_w P(S_1 = f(w)|w) P(X \neq f(w)|w, S_1 = f(w)) \\
= & \sum_w P(W = w, S_1 = f(w), X \neq f(w)) \\
= & \sum_w P_w d_w (1 - \beta_w) \\
= & \bar{\beta} d \qquad (30)
\end{aligned}$$

Using (23), (24), (25) and (28) yields the goal in (15).

Next, we show that the constraints are satisfied as follows.

$$\begin{aligned}
D & \geq \mathbb{E}\mathbf{d}(X, f(W)) \\
& = \mathbb{E}(X \oplus f(W)) \\
& = P(X \neq f(W)) \\
& = \sum_w P_w P(X \neq f(w)|W = w) \\
& = \sum_w P_w d_w \\
& = d, \qquad (31)
\end{aligned}$$

and

$$\begin{aligned}
p_1 & = P(X \neq S_1) \\
& = \sum_w P(X \neq S_1, W = w) \\
& = \sum_w [P(X = f(w), S_1 \neq f(w), W = w) \\
& \quad + P(X \neq f(w), S_1 = f(w), W = w)] \\
& = \sum_w [P_w (1 - d_w) \alpha_w + P_w d_w (1 - \beta_w)] \\
& = \alpha \bar{d} + \bar{\beta} d. \qquad (32)
\end{aligned}$$

Step 2: we prove that constraint $d \leq D$ can be met with equality without loss of optimality.

For given $D < 1/2$, suppose that the parameters that minimize the right hand side of (13) are (α^*, β^*, d^*) and furthermore $d^* < D$. We can a construct random variable W^* by using Table I with (α^*, β^*, d^*) replacing (α, β, d). Then we have

$$\widetilde{R}^{CR}(D) = I(X, S_1; W^*|S_2). \qquad (33)$$

Consider a random variable $W^{**} = W^* \oplus U$, where U is a Bernoulli random variable independent of all other variables with $P(U = 1) = \mu \in (0, 1/2)$, where parameter μ satisfies the constraint $d^* * \mu = D$, which is valid since $d^* < D < 1/2$. Clearly, $W^{**} \leftrightarrow W^* \leftrightarrow (X, S_1) \leftrightarrow S_2$ forms a Markov Chain and $E\mathbf{d}(X, W^{**}) = D$. From the definition of the rate distortion function for this problem, we can derive the upper bound of $\widetilde{R}^{CR}(D)$

$$I(X, S_1; W^{**}|S_2) \geq \widetilde{R}^{CR}(D). \qquad (34)$$

However, using the Markov chain $W^{**} \leftrightarrow W^* \leftrightarrow (X, S_1) \leftrightarrow S_2$ can yield the lower bound of $\widetilde{R}^{CR}(D)$ as follows,

$$\begin{aligned}
\widetilde{R}^{CR}(D) & = I(X, S_1; W^*|S_2) \\
& = I(X, S_1; W^*, W^{**}|S_2) \\
& \geq I(X, S_1; W^{**}|S_2) \qquad (35)
\end{aligned}$$

Combining (34) and (35) together yields the result $\widetilde{R}^{CR}(D) = I(X, S_1; W^{**}|S_2)$, from which we can conclude that constraint $d \leq D$ can be met with equality without loss of optimality.

Remark 1: For the case $D = 0$, we have

$$\widetilde{R}^{CR}(0) = H(X|S_2). \qquad (36)$$

Proof: Using $D = 0$ in the equality constraint of (11) yields $\alpha = p_1$. Setting parameters $\alpha = p_1$ and $D = 0$ into (11), we can compute $\widetilde{R}^{CR}(0)$ as follows,

$$\widetilde{R}^{CR}(0) = h(p_1) + G_{p_2}(p_1) = h(p_1 * p_2). \qquad (37)$$

Meanwhile, Using the assumptions in (8) yields

$$H(X|S_2) = H(X) + H(S_2|X) - H(S_2) = h(p_1 * p_2). \qquad (38)$$

Combing (37) and (38), we get the desired result.

Remark 2: For the case $p_1 = 0$, the model in Fig. 1 reduces to a Wyner-Ziv problem under CR constraint with parameter p_2, i.e., $S_1 = X, S_2 = X + N_2$, which implies that the encoder has no side information, and only the decoder has the side information S_2. We denote $\widetilde{R}^{CR}_{WZ}(D)$ as the rate distortion function for this model. By setting $p_1 = 0$ into (11), we have

$$\widetilde{R}^{CR}_{WZ}(D) = \begin{cases} h(p_2 * D) - h(D), & 0 \leq D < 1/2, \\ 0, & 1/2 \leq D \leq 1, \end{cases} \qquad (39)$$

where p_2 can be thought as the crossover probability of a binary symmetric channel (BSC) which connects the source X and the side information at the decoder.

Remark 3: For the case $p_2 = 0$, the model in Fig. 1 reduces to a conditional lossy coding problem under CR constraint with parameter p_1, i.e., $S_1 = S_2 = X + N_1$, which implies that the encoder and the decoder have the same side information S_2. We denote $\widetilde{R}^{CR}_{X|S_2}(D)$ as the rate distortion function for this model. We have

$$\widetilde{R}^{CR}_{X|S_2}(D) = R_{X|S_2}(D), \qquad (40)$$

where $R_{X|S_2}(D)$ is the conditional rate distortion function for the binary source in [11]. This result implies that the CR constraint has no impact on rate distortion performance when both the encoder and the decoder has the same side informatioin.

Proof: Recalling the function $R_{X|S_2}(D)$ in [11], we only need to verify

$$\widetilde{R}^{CR}_{X|S_2}(D) = \begin{cases} h(p_1) - h(D), & 0 \leq D < 1/2, \\ 0, & 1/2 \leq D \leq 1, \end{cases} \tag{41}$$

where p_1 can be thought as the crossover probability of a binary symmetric channel which connects the source X and the side information at the decoder.

Next, we only discuss the case $D < 1/2$. We firstly bound the function $F(\alpha, \beta, D)$, and show this bound can be achieved. Using $p_2 = 0$ yields

$$G_{p_2}(\alpha\overline{D} + \beta D) = h(\alpha\overline{D} + \beta D) - h(\alpha\overline{D} + \beta D) = 0. \tag{42}$$

Following from the concavity of the binary entropy function, we have

$$\begin{aligned} & (\overline{\alpha}\overline{D} + \overline{\beta}D)h(\frac{\overline{\beta}D}{\overline{\alpha}\overline{D} + \overline{\beta}D}) + (\alpha\overline{D} + \beta D)h(\frac{\beta D}{\alpha\overline{D} + \beta D}) \\ \leq\ & h\left((\overline{\alpha}\overline{D} + \overline{\beta}D)\frac{\overline{\beta}D}{\overline{\alpha}\overline{D} + \overline{\beta}D} + (\alpha\overline{D} + \beta D)\frac{\beta D}{\alpha\overline{D} + \beta D}\right) \\ =\ & h(D). \end{aligned} \tag{43}$$

Combing (42) and (43) into (10) and (11), the lower bound of $\widetilde{R}^{CR}_{X|S_2}(D)$ is derived as below,

$$\widetilde{R}^{CR}_{X|S_2}(D) \geq h(p_1) - h(D). \tag{44}$$

Note that this lower-bound can be achieved if we set $\alpha = \beta$ in (11). That is, when $D < 1/2$, we have

$$\widetilde{R}^{CR}_{X|S_2}(D) = h(p_1) - h(D). \tag{45}$$

Remark 4: Fig. 2 shows the curves of the rate distortion functions $\widetilde{R}^{CR}(D)$, $\widetilde{R}^{CR}_{WZ}(D)$, $\widetilde{R}^{CR}_{X|S_2}(D)$. In detail, the curve $\widetilde{R}^{CR}(D)$ in Fig. 1 is plotted with parameters $p_1 = 0.1,\ p_2 = 0.05$. Note that in Fig. 1, the crossover probability connecting the source and the decoder side information is $p_1 * p_2$, the curves $\widetilde{R}^{CR}_{WZ}(D)$ and $\widetilde{R}^{CR}_{X|S_2}(D)$ are plotted with the same crossover probability $p_1 * p_2 = 0.1 * 0.05 = 0.14$.

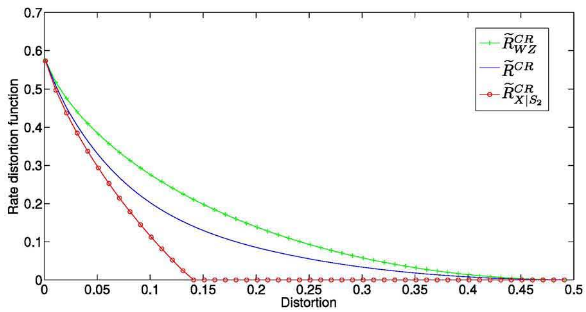

Fig. 2. The curves of the rate distortion functions $\widetilde{R}^{CR}(D)$ with parameters $p_1 = 0.1,\ p_2 = 0.05$, $\widetilde{R}^{CR}_{WZ}(D)$ and $\widetilde{R}^{CR}_{X|S_2}(D)$ with the same cross probability $p_1 * p_2 = 0.14$, which can be thought as the crossover probability of a binary symmetric channel that connects the source X and the decoder side information.

V. Appendix

Proof of Lemma 1:

$$\begin{aligned} & I(X, S_1; W|S_2) \\ \overset{(g)}{=}\ & I(X, S_1, S_2; W) - I(S_2; W) \\ \overset{(h)}{=}\ & I(X, S_1; W) - I(S_2; W) \\ =\ & H(X, S_1) - H(X, S_1|W) - [H(S_2) - H(S_2|W)] \\ =\ & H(X, S_1) - H(S_1|W) - H(X|W, S_1) \\ & -H(S_2) + H(S_2|W), \end{aligned}$$

where (g) follows from the chain rule for mutual information, (h) from the condition that $W \leftrightarrow (X, S_1) \leftrightarrow S_2$ is a Markov chain.

VI. Conclusion

We have considered the lossy coding problem with encoder side information and common reconstruction constraint. In this scenario, the rate distortion function is calculated explicitly for the doubly-symmetric binary source.

VII. Appendix

Acknowledgment

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References

[1] A. D. Wyner and J. Ziv, "The rate-distortion function for source coding with side information at the decoder," *Information Theory, IEEE Transactions on*, vol. 22, no. 1, pp. 1–10, 1976.

[2] R. M. Gray, *Conditional rate-distortion theory*. Information Systems Laboratory, Stanford Electronics Laboratories, 1972.

[3] ——, "A new class of lower bounds to information rates of stationary sources via conditional rate-distortion functions," *Information Theory, IEEE Transactions on*, vol. 19, no. 4, pp. 480–489, 1973.

[4] B. Guler, E. MolavianJazi, and A. Yener, "Remote source coding with two-sided information," in *Information Theory (ISIT), 2015 IEEE International Symposium on*. IEEE, 2015, pp. 2176–2180.

[5] T. Laich and M. Wigger, "Utility of encoder side information for the lossless kaspi/heegard-b erger problem," in *Information Theory Proceedings (ISIT), 2013 IEEE International Symposium on*. IEEE, 2013, pp. 3065–3069.

[6] Y. Steinberg, "Coding and common reconstruction," *Information Theory, IEEE Transactions on*, vol. 55, no. 11, pp. 4995–5010, 2009.

[7] A. Lapidoth, A. Malar, and M. Wigger, "Constrained source-coding with side information," *Information Theory, IEEE Transactions on*, vol. 60, no. 6, pp. 3218–3237, 2014.

[8] B. N. Vellambi and R. Timo, "Successive refinement with common receiver reconstructions," in *Information Theory (ISIT), 2014 IEEE International Symposium on*. IEEE, 2014, pp. 2664–2668.

[9] ——, "The heegard-berger problem with common receiver reconstructions," in *Information Theory Workshop (ITW), 2013 IEEE*. IEEE, 2013, pp. 1–5.

[10] J. Lu, Y. Xu, Y. Qian, and Q. Wang, "The heegard-berger problem with encoder side information and common reconstruction constraints for the binary source," *in prepare*.

[11] T. Berger, *Rate distortion theory: A mathematical basis for data compression*. Englewood Cliffs, N.J.: Prentice-Hall, 1971.